FEUP

FACULTY OF ENGINEERING - UNIVERSITY OF PORTO

Systems Engineering

Market Survey

SWAID - Plate Resonance (PR)

PLATE RESONANCE
TEAM

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1 Introduction

1.1 System Overview

The Standing Wave Interactive Demonstrator (SWAID) project aims to develop an interactive system capable of visualizing standing wave patterns on a vibrating plate through the formation of **Chladni patterns**. These patterns emerge when a rigid plate is excited at specific frequencies, causing fine particles (such as sand or powder) to accumulate along the nodal lines of the vibration modes.

The proposed system integrates multiple subsystems responsible for signal generation, mechanical excitation, visual enhancement, and user interaction. By controlling parameters such as excitation frequency, phase, and amplitude, the system will be capable of producing a wide range of vibration modes on the plate surface.

The final objective is to build a platform that can clearly demonstrate acoustic resonance phenomena while allowing users to interact with the generated patterns.

1.2 Objectives

The goals of this market survey are:

- Identify and analyse technologies available for synchronized acoustic signal generation.
- Analyse interactive installations using sound and vibration.
- Identify design approaches and components commonly used in acoustic visualization systems.

1.3 Terms and Definitions

SWAID Standing Wave Acoustic Interactive Demonstrator.

PR Plate Resonance (Plate Resonance Subsystem).

FPGA Field-Programmable Gate Array.

DAC Digital-to-Analog Converter.

Chladni Pattern Geometric pattern formed by particles on a vibrating plate when excited at resonance frequencies.

DIY Do it yourself.

1.4 Changes

Version	Date	Author	Description
1	27th Feb	Lucas Aparício	Template creation
2	12th Mar	João Pinto, Beatriz Martins	First Version

2 Methodology

2.1 Market Segmentation

The target market for SWAID-type systems can be divided into four main categories, based on user profile and application objectives.

- **Edutainment:** Focuses on museums and science centers where the SWAID system must be ”plug-and-play” and durable enough for public use.
- **Multimedia and Art Installations:** Artists and content creators require systems that allow easy audiovisual synchronization, strong software integration, and scalability for larger installations.
- **Educational Institutions:** Schools and universities that use equipment to demonstrate and engage students in the physics of waves and resonance. These systems require clarity in demonstrations, reliability, and ease of use.
- **Industrial Acoustics:** Specialized applications such as modal analysis, non-destructive material testing, and advanced vibration studies. These systems require deterministic real-time behaviour and high levels of parallel signal control.

2.2 Relevant Technologies

We evaluated two primary hardware paths for the SWAID-PR subsystem:

- **FPGA + Custom Drivers:** Utilizes high-speed logic for Direct Digital Synthesis (DDS). While this approach allows for a virtually unlimited and flexible number of output channels with near-zero latency, it introduces significant development complexity and PCB fabrication costs.
- **Multichannel Audio Interface:** Standard 5.1 or 7.1 sound cards. These provide 6 to 8 high-fidelity output channels. By utilizing low-level **ALSA (Advanced Linux Sound Architecture)** drivers, software latency can be minimized to levels appropriate for real-time interaction.

Team Decision: The team has concluded that the **Multichannel Audio Interface** is the most viable solution. Although the FPGA offers superior channel scalability, the physical dimensions and mechanical properties of the Chladni plate restrict the practical number of transducers to between 4 and 6. Beyond this range, transducer interference and hardware costs increase disproportionately without providing significant benefit to pattern diversity.

Since a standard 5.1 or 7.1 sound card natively supports this requirement, the channel leverage of an FPGA is unjustified. By choosing the audio interface and optimizing via ALSA drivers, we significantly reduce project risk and complexity while maintaining the necessary real-time response.

2.3 Evaluation Criteria

The available technologies were evaluated according to the following criteria:

- Hardware Cost
- Development Time and Risk
- Integration Complexity
- Response Latency

- Signal Fidelity

2.4 Information Sources

The data presented in this market survey was obtained from publicly available technical documentation, manufacturer specifications, and academic demonstrations related to acoustic excitation systems and Chladni pattern visualization.

Sources consulted include:

- Manufacturer documentation for signal generators and audio interfaces, such as the *CLS Computer* and *Creative Labs* multichannel audio interface product lines.
- FPGA development board documentation from manufacturers such as *Lattice* and *Sipeed*, commonly used for (low-cost) digital signal processing and Direct Digital Synthesis (DDS) implementations.
- Educational physics demonstrations of Chladni plates from university laboratory experiments and physics teaching material.
- Online open-source projects related to acoustic visualization and Chladni pattern generation, including experimental implementations shared through platforms such as GitHub.
- Technical tutorials and experimental demonstrations available through scientific outreach channels and educational resources focusing on acoustic resonance and standing wave visualization.

2.5 Information Gathering and Organization

Data collection focused on technologies that are feasible for the selected market segments. Since Chladni patterns can be produced within the normal audible frequency range (below 20 kHz), the research focused on evaluating whether standard audio equipment could be used to generate the required excitation signals.

Hardware costs, performance characteristics, and latency metrics were analysed and organized into comparison tables to support a balanced evaluation of different technological approaches while minimizing implementation risks.

3 Comparative Tables

The following table summarizes the existing technological approaches. The comparison highlights why the sound card approach is the balanced choice for our specific physical constraints.

Table 1: Comparison of Existing Acoustic Excitation Systems

System Type	Signal Gen.	Channels	Latency	Complexity	Technical Note
Traditional Demonstrator	Analog Gen.	1	N/A	Low	Limited to single-frequency study; no multi-point control.
Audio Interface	**Sound Card**	**6–8**	**Low***	**Low**	**Selected: 5.1/7.1 is sufficient for 4–6 transducers via ALSA.**
FPGA Generator	DDS Logic	8+	Ultra-Low	High	Over-engineered for plate size; high development time/cost.
Microcontroller	PWM / DAC	1–2	Moderate	Mod/High	Difficult to synchronize phase,amplitude and frequency across multiple channels.

* *Low latency achieved through direct ALSA driver optimization.*